The examination is being carried out on the **following application documents**

Description, Pages

1-54 as published

Sequence listings, SEQ ID NO

1-5 as published

Claims, Numbers

1-15 filed in electronic form on 21-02-2024

Drawings, Sheets

1/7-7/7 as published

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 MING-TZO WEI ET AL: "Nucleated Transcriptional Condensates Amplify Gene Expression",
BIORXIV, 21 August 2019 (2019-08-21), pages 1-20, XP055694863,
DOI: 10.1101/737387

D2 PESSINA FABIO ET AL: "Functional transcription promoters at DNA double-strand breaks mediate RNA-driven phase separation of damage-response factors",
NATURE CELL BIOLOGY, NATURE PUBLISHING GROUP UK, LONDON, vol. 21, no. 10, 30 September 2019 (2019-09-30), pages 1286-1299, XP036895332,
ISSN: 1465-7392, DOI: 10.1038/S41556-019-0392-4
[retrieved on 2019-09-30]

The following documents have been cited in the international search report; the numbering will be adhered to in the rest of the procedure.

D3 WO 2020/232416 A1 (WHITEHEAD INSTITUTE FOR BIOMEDICAL RESEARCH) 19 November 2020 (2020-11-19)

D4 Song Daesun, Jo Yongsang, Choi Jeong-Mo, Jung Yongwon: "Client proximity enhancement inside cellular membrane-less compartments governed by client-compartment interactions", Nature Communications, vol. 11, no. 1, 1 December 2020 (2020-12-01), XP055966919, DOI: 10.1038/s41467-020-19476-4

1 The amendment filed with the letter dated 21-02-2024 introduces subject-matter which extends beyond the content of the application as filed, contrary to Article 123(2) EPC. The amendment concerned is the following:

1.1 The subject-matter of present claim 2 finds a partial basis in claim 2 of the application as filed, hereinafter referred to as the "PCT application". However, present claim 2 has been further amended to depend on present claim 1, whereas it was an independent claim in the PCT application. This leads to added subject-matter for which there is no appropriate basis in the application. After all, claim 2 PCT did not contain all the features of claim 1 PCT (which corresponds to present claim 1), so that it cannot be rendered dependent thereon without violating Art. 123(2) EPC. In particular, in the last determination step of claim 1 a comparison is made with a change in an intensity and a size of membrane-less bodies in the absence of the agent in question to (as a conclusion) indicate that said agent disrupts binding of said target peptide to said biomolecule. In claim 2, on the other hand the determination step is relative as regards such a change for a control peptide. In other words the change per Se as regards intensity and a size of membrane-less bodies in the absence of the agent for the first cell in claim 2 PCT does not indicate the same conclusion as in claim 1. As a consequence claim 2 contains added subject-matter, because now it does require that it indicates the conclusion of claim 1 PCT.

2 The application does not meet the requirements of Articles 84 and 83 EPC, because claims 1-15 are neither clear, nor technically supported nor sufficiently disclosed over the whole scope claimed.

2.1 In the first place, the claims require that membrane-less bodies comprising said target molecule are detected, wherein said detecting comprises measuring a change in an intensity and a size of said membrane-less bodies. The application provides technical support and disclosure only for one way to enable this detection of intensity and size, namely by using fluorescence detection as in dependent claim 10, by virtue of the target peptide being fused to a fluorophore as in present dependent claim 3. As claims 3 and 10 are mere dependent claims, the other claims not necessarily having these features are supposedly envisaged to be able to achieve this detection of intensity and size of the

membrane-less bodies in an alternative way. The problem is that no such alternative way is disclosed in the application. Therefore the detection of intensity and size, by using fluorescence detection, by virtue of the target peptide being fused to a fluorophore from a technical point of view appear to be essential features (Guidelines F-Iv, 4.5), even if the application presents them as mere optional features. The applicant is is requested to either limit to these features or explain what other alternatives the skilled person would immediately know/envisage from their common knowledge. The absence of these features in most claims render the claims unclear and technically unsupported.

2.2 It is also not understood what "an intensity" in the claims is supposed to encompass other than fluorescence intensity. The term is therefore not clear.

2.3 From the examples (in particular par.0156) but also from par.0042 it becomes clear that only if an increase in the change of fluorescence intensity and a size of said membrane-less bodies is detected upon addition of the agent, as regards the same criteria in the absence of the agent, it can be concluded/ determined that the binding of the target peptide to the biomolecule is disrupted. After all this is the only interpretation that makes technical sense: If the fluorescent target peptide is incorporated into the membrane-less bodies when said target peptide is released from the binding to the biomolecule and undergoes phase separation at a sub-region of said cell that is separate and distinct from a sub-region where said target peptide binds to said biomolecule, the membrane-less bodies increase in size and gain fluorescence intensity. It is not understood how this incorporation of the target molecule into said membrane-less bodies could possibly lead to a decrease. However, if one simply follows the technical instructions of the claims (i.e. contacting a cell with a putative exogenous agent, wherein said cell comprises a target molecule that comprises said target peptide and detecting by measuring a change in an intensity and a size of said membrane-less bodies), in the case that the change is a decrease, one may be detecting exogenous agents that do have an effect on phase separation, but do not disrupt binding of a target peptide to a biomolecule. This is the case for document D2 (fig 7 and 8 + legends and page 1294-1296, section:"53BP1 foci at DSBs show LLPS characteristics" as well as page 1296, sections: "RNA synthesis drives phase separation of 53BP1 at DSB in vitro" and "Phase separation and de novo transcription impact DSB repair"): addition of the exogenous agent THZ leads to less foci (e.g. fig 8c), not because the binding between target peptide 53BP1 fused to GFP and the biomolecule DDRNA1 is disrupted, but because THZ inhibits transcription and hence leads to decreased 53BP1-GFP in the membrane-less bodies.

2.4 The features "an exogenous agent that disrupts binding of a target peptide to a biomolecule" and "wherein said target molecule is incorporated into said membrane-less bodies when said target peptide is released from said binding to said biomolecule and undergoes phase separation at a subregion of said cell that is separate and distinct from a sub-region where said target peptide binds to said biomolecule" in conjunction are neither clear, nor technically supported nor sufficiently disclosed over the whole scope claimed. Not just for any target peptide that interacts with a biomolecule, disruption of this interaction will necessarily lead to incorporation of the target molecule into said membrane-less bodies when said target peptide is released from said binding to said biomolecule and undergoes phase separation as claimed.

In this respect, D1-D3 present verifiable facts which cast serious doubts on the enablement of the claimed subject-matter over the whole scope claimed. For D2, see item 2.3 above: by carrying out exactly the same steps using entities that appear to fall in the claimed scope, one does not arrive at a solution to the problem: no agent which disrupts binding of a target peptide to a biomolecule is identified, rather an agent that inhibits transcription is identified. This means that the claimed concept is not fit for generalisation.

In D3 (see pars.0058, 0059 and figures 24 and 25) the GFP labeled target peptide MED binds to the estrogen receptor (ER). Upon addition of the exogenous FXXFF-type agent of SEQ ID No 1 which disrupts this interaction, it is not so that now said target peptide is released from said binding to said biomolecule and undergoes phase separation at a subregion of said cell that is separate and distinct from a sub-region where said target peptide binds to said biomolecule. Rather, in this case the GFP-MED...ER complex was already present in a phase separated droplet, and disruption of the interaction leads to the effect that ER is cast out from the droplet. Although it is not indicated what the effect of this is in terms of fluorescence intensity and size of the particles, it is to be expected that the size will decrease due to the loss of the ER molecules, but the intensity will increase as the same amount of GFP molecules are now present in a smaller droplet volume. Thus by carrying out exactly the same steps using entities that appear to fall in the claimed scope, one does not arrive at a solution to the problem. This means that the claimed concept is not fit for generalisation.

Even if the claims were to be limited to the detection of an increase in the change of fluorescence intensity and a size of said membrane-less bodies, then still this concept cannot be considered to have been sufficiently disclosed over the whole scope claimed. Even if the target peptide comprises an IDR, like

TAF15 as in present claims 5 and 13, it is still not warranted that carrying out the steps as claimed lead to the claimed effect over the whole scope claimed. In D1 (pages 3-4, section: "Transcriptional condensates colocalize with and recruit unphosphorylated CTD") a labelled variant of TAF15 (optoTAF15) as target molecule comprising a target peptide binds to RNA polymerase II as biomolecule. Upon addition of the exogenous agent THZ1, the phase separation increases. This would all appear to follow exactly what claim 1 requires. Except that this does not mean that now an inhibitory activity of an exogenous agent that disrupts binding of a target peptide to a biomolecule has been identified. Rather, due to the fact that THZ1 inhibits the kinase CDK7, less CTD moieties of the RNA pol II are phosphorylated. Because unphosphorylated RNA pol II interacts more strongly with optoTAF1, this promotes the phase separation and increases the size intensity of the membrane-less bodies (figures 2-4, in particular 3E, 3G, 4A and 4C; pages 3-4 section: "Transcriptional condensates colocalize with and recruit unphosphorylated CTD"). Thus by carrying out exactly the same steps using entities that appear to fall in the claimed scope, one does not arrive at a solution to the problem. This also means that the claimed concept is not fit for generalisation.

In particular, the broad wording presently used involves an undue burden of screening to find suitable combinations of target peptides and biomolecules for which the target molecule is incorporated into said membrane-less bodies when said target peptide is released from said binding to said biomolecule and undergoes phase separation at a sub-region of said cell that is separate and distinct from a sub-region where said target peptide binds to said biomolecule.

Sufficiency of disclosure and technical support can only be recognized for the *de facto* disclosed example 2. Examples 1 and 3 are mere prophetic examples, that in view of D1-D3 have no value of evidence. In example 2 the target peptide is BRD4 and the biomolecule it interacts with is a histone. In that case the exogenous agent JQ1 is able to disrupt the binding between the fluorophore labelled BRD4 and histone upon which BRD4 undergoes phase separation at a sub-region of said cell that is separate and distinct from a sub-region where said target peptide binds to said biomolecule and results in a measured increase in an intensity and a size of said membrane-less bodies relative to a change in an intensity and a size of membrane-less bodies in the absence of said agent indicates that said agent disrupts binding of said target peptide to said biomolecule. If any subject-matter may be found patentable, it must reflect these essential components of example 2.

3 The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of independent claim 1 does not involve an inventive step within the meaning of Article 56 EPC.

3.1 D3 (see pars.0058, 0059 and figures 24 and 25) can be considered as the closest state of the art and discloses a method of identifying an inhibitory activity of an exogenous agent (e.g. FXXFF-type agent of SEQ ID No 1) that disrupts binding of a target peptide MED to a biomolecule ER, the method comprising:
(i) contacting a cell with said agent, wherein said cell comprises a target molecule that comprises said target peptide;
(ii) detecting membrane-less bodies comprising said target molecule,
wherein said detecting comprises measuring a change in an intensity and a size of said membrane-less bodies; and wherein a bigger change in said intensity and said size of said membrane-less bodies relative to a change in an intensity and a size of membrane-less bodies in the absence of said agent indicates that said agent disrupts binding of said target peptide to said biomolecule.

3.2 Claim 1 differs therefrom in that 1) the target molecule must be incorporated into said membrane-less bodies when said target peptide is released from said binding to said biomolecule and undergoes phase separation at a sub-region of said cell that is separate and distinct from a sub-region where said target peptide binds to said biomolecule and 2) detecting comprises measuring a change in an intensity and a size of said membrane-less bodies; and wherein a bigger change in said intensity and said size of said membrane-less bodies relative to a change in an intensity and a size of membrane-less bodies in the absence of said agent indicates that said agent disrupts binding of said target peptide to said biomolecule.

3.3 The presence of an inventive step cannot be recognized over the whole scope claimed for the following reasons.

A *conditio sine qua non* for the recognition of inventive step (Article 56 EPC) of a claim is that a solution to the problem posed in the claim is provided. If the inventive step of a claimed invention is based on a given technical effect, the latter should, in principle, be achievable over the whole area claimed (T939/92). As explained under items 2.1, 2.3 and 2.4 above, this is not the case here since the patent specification does not demonstrate the achievement of the desired technical effect of identifying an inhibitory activity of an exogenous agent that disrupts binding of a target peptide to a biomolecule over the whole scope of the claims, inventive step must be denied.

4 Dependent claims

4.1 Dependent claims 2-15 as presently claimed do not contain any features which, in combination with the features of any claim to which they refer, meet the requirements of the EPC in respect of inventive step, the reasons being as follows:

The incorporation of the features of claims 2-15 into the corresponding independent claims alone does not remedy the deficiency under item 3.3. To remedy this deficiency all the essential features of example 2 would need to be incorporated in a manner that does not offend Art.123(2) EPC.

5 The applicant is requested to file new claims which take account of the above comments.

5.1 In order to facilitate the examination of the conformity of the amended application with the requirements of Article 123(2) EPC, the applicant is requested to clearly identify the amendments carried out, irrespective of whether they concern amendments by addition, replacement or deletion, and to indicate the passages of the application as filed on which these amendments are based.

5.2 When filing amended claims the applicant should at the same time bring the description into conformity with the amended claims. Care should be taken during revision, especially of the introductory portion and any statements of problem or advantage, not to add subject-matter which extends beyond the content of the application as originally filed (Article 123(2) EPC). Subject-matter no longer claimed (excess subject-matter) should be deleted or be made clear not to be part of the "invention" or an "embodiment" thereof.

5.3 The phrase "incorporated herein by reference" on several pages of the description should be deleted. According to the Guidelines, F-III, 8, the patent specification should be self contained i.e. capable of being understood without reference to any other document.

5.4 References to unpublished US applications must be replaced with the corresponding published US applications.

5.5 References to gametes and embryos in pars.0082 and 0086 of the description will need a "non-human"-disclaimer in view of Art. 53(a) EPC in conjunction with Rule 29 EPC.

5.6 To meet the requirements of Rule 42(1)(b) EPC, the documents D1-D4 should be identified in the description and the relevant background art disclosed therein should be briefly discussed.